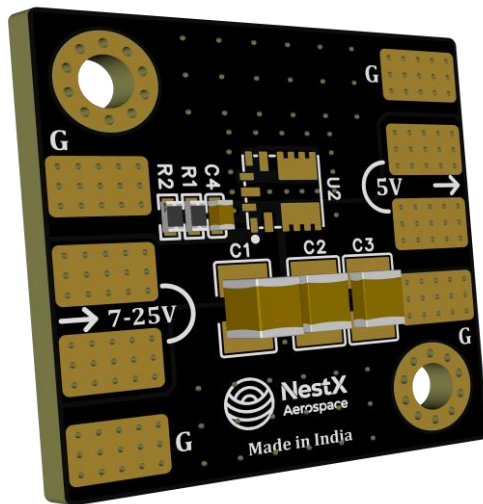


NX-Joule 1

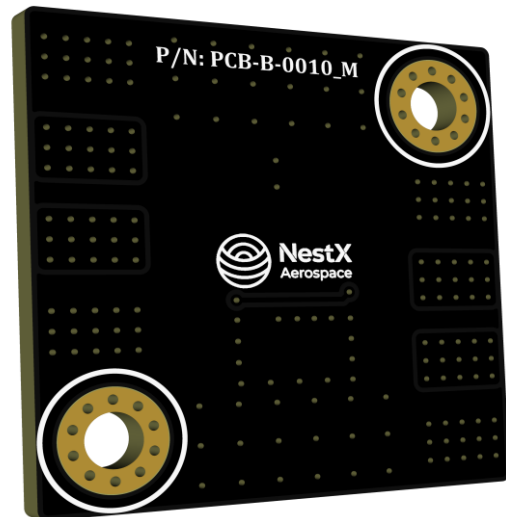
2S – 6S Wide-Input Buck Converter Module

Regulated 5V / 3.3V Output | Up to 2.5A | For Unmanned Aerial & Autonomous Systems | **Made in India**

Product Images



Top View



Bottom View

General Description

The NX-BUCK Series is a compact, high-efficiency synchronous buck converter module designed for unmanned aerial, satellite, and embedded aerospace applications. Accepting a wide 2S–6S LiPo / LiHV battery input (7V–25V), it delivers regulated 5V or 3.3V output at up to 2.5A continuous current (with heatsink). The module is built around an industrial-grade IC with full protection features, rigorous EMI filtering, and a smd pad-based interface.

Key Features

- ✓ Wide input range: 7V to 25V (2S–6S LiPo / LiHV)
- ✓ Fixed output options: 5.0V or 3.3V
- ✓ Output current: up to 1.5A (no heatsink) / 2.5A (with heatsink)
- ✓ Overcurrent Protection — Hiccup Mode auto-recovery
- ✓ Over-Temperature Thermal Shutdown Protection
- ✓ Meets EN55011 Radiated EMI Standards
- ✓ Extra distribution pads for daisy-chain power distribution
- ✓ M2.5 mounting holes for secure mechanical attachment
- ✓ Ultra-compact: 25 × 25 mm footprint, 10 g weight
- ✓ Operating IC Junction: –40°C to +125°C | Ambient: –40°C to +85°C
- ✓ Made in India — space-heritage manufacturing

Electrical Specifications

Absolute Maximum Ratings ($T_A = 25^\circ\text{C}$ unless noted)

Parameter	Min	Typ	Max	Unit
Input Voltage (V_{IN})	7	—	28	V
Output Voltage — 5V Option (V_{OUT})	4.90	5.00	5.10	V
Output Voltage — 3.3V Option (V_{OUT})	3.24	3.30	3.36	V
Continuous Output Current (no heatsink)	—	—	1.5	A
Continuous Output Current (with heatsink)	—	—	2.5	A
Switching Frequency	—	750	—	kHz
Efficiency ($V_{OUT} = 5\text{V}$, $I_{OUT} = 1\text{A}$)	—	88	—	%
Efficiency ($V_{OUT} = 3.3\text{V}$, $I_{OUT} = 1\text{A}$)	—	83	—	%
Short-Circuit Protection Recovery (Hiccup)	—	Auto	—	—
Thermal Shutdown Threshold (T_J)	—	165	—	$^\circ\text{C}$

Thermal & Operating Conditions

Parameter	Min	Typ	Max	Unit
IC Junction Temperature (T_J)	-40	—	+125	$^\circ\text{C}$
Operating Ambient Temperature (T_A)	-40	—	+85	$^\circ\text{C}$

Efficiency vs. Output Current

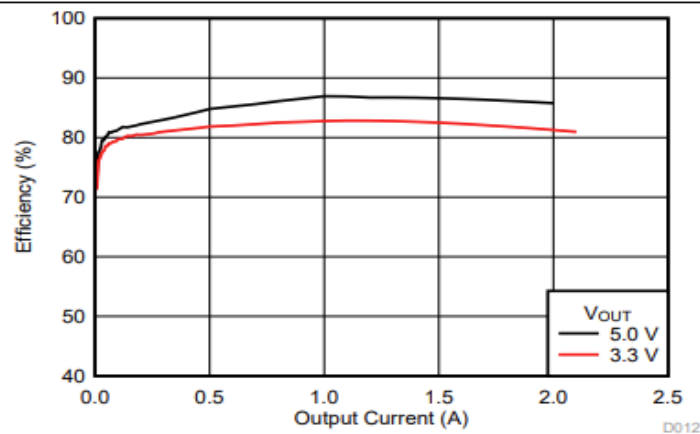


Figure 1 — Efficiency vs. Output Current ($V_{IN} = 12\text{V}$)

Mechanical Specifications

PCB Dimensions	25.0 mm × 25.0 mm
Board Thickness	3.2 mm (standard)
Weight	10 g (without heatsink)
Mounting	M2.5 threaded standoffs (×2 diagonal)
Connector Interface	Flat SMD copper pads
PCB Color	Black ENIG finish (or as per requirement)

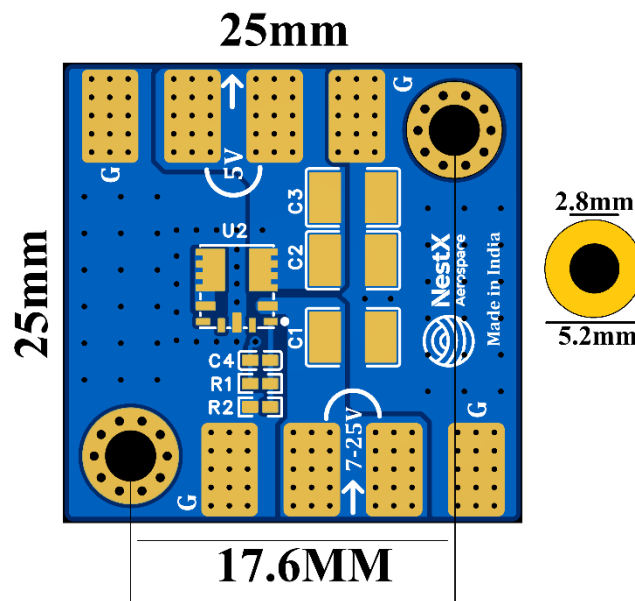
Compliance & Standards

EMI Radiated Emissions	Meets EN55011 Class B
Operating Temperature (IC)	−40°C to +125°C (Junction)
Operating Temperature (Ambient)	−40°C to +85°C
Storage Temperature	−55°C to +150°C
Country of Origin	India

Pin / Pad Description

Label	Type	Description
7-25V	Power Input	Primary voltage input (2S-6S battery). Two parallel pads distribution.
5V / 3V	Power Output	Regulated 5V/3V output rail. Two parallel pads distribution.
G	GND	Ground reference. Multiple pads distributed around board perimeter for low-impedance return.
DIST. PADS	Distribution	Extra solder pads for distributing power to additional boards.

Dimensions:



Application Notes

1. Ensure adequate airflow or thermal path when running at maximum current (2.5A).
2. Ground pads must be soldered completely on all perimeter pads for optimal EMI and thermal performance.
3. The M2.5 mounting holes are ground-connected; use non-conductive standoffs if isolation is required.

Ordering Information

Part Number	Output Voltage	Max Current	Notes
NX01-01-5.0	5.0 V	1.5 A	No heatsink
NX01-01-5.0-HS	5.0 V	2.5 A	With heatsink
NX01-01-3.3	3.3 V	1.5 A	No heatsink
NX01-01-3.3-HS	3.3 V	2.5 A	With heatsink

Custom output voltages and bulk pricing available on request.

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